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ORIGINAL ARTICLE

Holistic metabonomic profiling of urine affords potential early diagnosis for bladder and kidney cancers

Zhenzhen Huang · Yongjing Chen · Wei Hang · Yao Gao · Lin Lin · Daniel Y. Li · Jinchun Xing · Xiaomei Yan

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Abstract The clinical exploration of urinary metabonomic analysis on discriminating between the top-two-incidence urological cancers, bladder and kidney cancers (BC and KC), is still virgin land. In this study, we discovered and evaluated the clinical utility of holistic metabonomic profiling and current single biomarker methods, and ultimately suggested a potential screening test for BC and KC. Urine metabonomic profiling for 19 BC patients, 25 KC patients, and 24 healthy controls was carried out using an LC–MS based platform, which utilized both reversed phase chromatography and hydrophilic interaction chromatography. The holistic method that refers to orthogonal partial least-squares-discriminant analysis based on all qualified profile data, successfully classified BC, KC and healthy control groups, showing 100 % sensitivity and specificity. Taurine, hippuric acid, phenylacetylglutamine and carnitine species contributed most to the BC and KC discrimination. The predictive power of each above metabolite is evaluated using receiver operator characteristic technique. Hippuric acid was found 10-fold decrease in concentration relative healthy controls, and performed the best as a biomarker for BC diagnosis, with its sensitivity and specificity of 78.9 and 86.5 %, respectively. Carnitine C10:3 was found 1.5-fold decrease, and reached 84.0 % of sensitivity and 60.5 % of specificity for KC diagnosis. In view of both sensitivity and specificity, the holistic method is more accurate for detecting BC and KC, than current single metabonomic biomarker methods, and it could be advocated as a prescreen to other forms of more invasive or uncomfortable screening. Moreover, this approach also demonstrates attractive performance in diagnosis of early stage (ES) BC and KC patients.

Keywords Metabonomics · Urine · Kidney cancer · Bladder cancer · Early diagnosis

Abbreviations

BC	Bladder cancer
KC	Kidney cancer
RPLC	Reversed phase liquid chromatography
HILIC	Hydrophilic interaction chromatography
TNM	Tumor nodes metastasis
ESI	Electrospray ionization
TIC	Total ion chromatogram
PCA	Principal component analysis
OPLS-DA	Orthogonal partial least-squares-discriminant analysis
ES	Early stage
HA	Hippuric acid
PAGN	Phenylacetylglutamine

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Z. Huang · Y. Chen · W. Hang (✉) · Y. Gao · L. Lin · X. Yan
Department of Chemistry, Key Laboratory of Analytical Sciences, College of Chemistry and Chemical Engineering, Xiamen University, 422 Simingnan Road, Xiamen 361005, China
e-mail: weihang@xmu.edu.cn

W. Hang
State Key Laboratory of Marine Environmental Science, Xiamen University, Xiamen, China

D. Y. Li
Department of Medicine, University of California, San Diego, CA, USA

J. Xing
Department of Urology, Xiamen First Hospital, Xiamen, China

ROC	Receiver operating characteristic
QC	Quality control
VIP	Variable importance in the project
AUC	Area under the curve

1 Introduction

Kidney cancer (KC) attacks approximately 208,000 people in the world every year (Linehan et al. 2003). Generally, it can be cured (survival rate >70 %) if it is diagnosed and treated when still localized to the kidney and surrounding tissue (National Cancer Institute, http://www.cancer.gov; Cohen and McGovern 2005). Unfortunately, more than 50 % of the patients exhibit locally advanced- or metastatic-KC at the time of diagnosis, and a poor prognosis (5-year survival rate <2 %) (Kutikov et al. 2010; Kim et al. 2009). Bladder cancer (BC) is the second most common genitourinary malignancy (Siegel et al. 2011). Approximately 70 % BC patients can acquire long-term survival, associated with superficial papillary lesions limited to the mucosa and submucosa of the bladder (Mohammed and Rahman 2008). Screening methods, preferably noninvasive, that facilitate early diagnosis of BC and KC would be advantageous for survival. However, for KC, there is no specific diagnostic modality other than incidental radiologic discovery (computed tomography, ultrasonography, or magnetic resonance imaging) (Morrissey et al. 2010; Michael and Pandha 2003). For BC, cystoscopy is invasive, while urinary cytology cannot achieve high accuracy for early stage (ES) cancer detection (Ott et al. 2009).

In recent years, BC and KC diagnoses have been approached using metabonomic analysis of urine (Kim et al. 2009; Morrissey et al. 2010; Mitra and Cote 2010). The metabolome is considered as the third pillar of system biology, after genome and proteome, and can be viewed as the end of the biochemical cascade (Theodoridis et al. 2011; Nicholson et al. 1999; Quackenbush 2006). Urinary metabonomics has several major advantages including the ready availability, the relative ease of urine analysis, and the fact that the derived metabonomic profiles are sensitive to both environmental and genomic changes affecting the pathogenesis and progression of diseases (Claudino et al. 2007; Pasikanti et al. 2010). Small molecule metabolites associated with BC and KC are freely filtered into urine because both bladder and kidney are directly contact to it. Thus, there is a need to determine how BC and KC influence urine, and whether the urine change can result in a discrimination between BC and KC.

There were several studies pinpointing single compound in the urine as biomarker of BC and KC (Kind et al. 2007; Paik et al. 2010; Pasikanti et al. 2010). Unfortunately, no accepted metabonomic biomarkers that could be applied in clinical diagnosis were reported hitherto. One- or two-component chemical tests may provide a quick way to monitor health status. However, BC and KC are complex diseases involving systemic deregulation of cell proliferation, survival, apoptosis, and cell cycle (Slupsky et al. 2010; Evan and Vousden 2001). Therefore, in this study, we aimed at examining the endogenous metabolites with a holistic perspective to determine, firstly, whether urinary metabonomic profiling, rather than just one- or two-compound methods, can result in the potential diagnosis of BC and KC, secondly, whether the profiling has the early diagnosis ability, and thirdly, whether the profiling can discriminate the top-two-incidence genitourinary cancers (BC and KC) from each other.

2 Materials and methods

2.1 Urine collection and storage

19 BC patients, 25 KC patients, and 24 healthy controls from Xiamen First Hospital were enrolled in this study. All the patients were diagnosed by histopathology examination, and none had received chemotherapy or radiation before urine collection. Written informed consent was obtained from each subject before sample collection, and the institutional ethics committees approved the protocols outlined in our text. Detailed information of the clinicopathological characteristics of the tumor patients and healthy controls is provided in Table 1. Tumor stages were established according to the 2002 Tumor nodes metastasis (TNM) staging system (Greene et al. 2002). T1-2 without metastases is considered ES and T3-4 with metastases is considered advanced (Table 1) (Lin et al. 2011). All urine samples were collected in the morning before breakfast and stored at −80 °C until analysis.

2.2 Chemicals

HPLC-grade acetonitrile (ACN) was obtained from Tedia (Fairfield, OH, USA). HPLC-grade formic acid (FA) was purchased from Sigma/Fluka (Switzerland). Distilled water (18.2 MΩ) for chromatographic separation was prepared using a Milli-Q water purification system (Millipore, Billerica, USA). Ammonium acetate was purchased from Tedia (USA). The standard hippuric acid (HA) and phenylacetylglutamine (PAGN) was purchased from Ma Jin (China).

2.3 Urine preparation

Urine samples were thawed at room temperature before analysis. For RPLC separation, a volume of 800 μL

Table 1 Demographic and clinical chemistry characteristics data

Characteristics	BC patients	KC patients	Healthy controls
No. of subjects	19	25	24
Age (mean, range)	60.5, 45–74	54.6, 27–71	50.5, 26–65
Male	14 (73.7 %)	15 (60 %)	15 (62.5 %)
Female	5 (26.3 %)	10 (40 %)	9 (37.5 %)
Cancer stage			
ES	14 (73.7 %)	9 (36 %)	\
T1N0M0	6	5	
T2N0M0	8	4	
Advanced-stage	5 (26.3 %)	16 (64 %)	\
T3aN0M0	0	3	
T3bN0M0	1	1	
T3aN1M0	2	1	
T3bN1M1	0	4	
T4bN3M1	1	2	
T4bN2M0	1	3	
T4bN2M1	0	2	
BMI	19.8, 16.2–24.3	21.2, 17.7–24.9	22.9, 18.2–26.8
Hematuria	2 (10.5 %)	4 (16 %)	0
Medications	0	0	0
Smoking habit			
Non smokers	6 (31.6 %)	14 (56 %)	16 (66.7 %)
Ex-smokers	0	3 (12 %)	0
Smokers	13 (68.4 %)	8 (32 %)	8 (33.3 %)
Race	Chinese	Chinese	Chinese

distilled water was added to 200 μL urine and then centrifuged at 13,000 rpm for 10 min at 4 °C. For HILIC separation, a volume of 800 μL of ACN was added to 200 μL urine. After vortexing, the mixture was set aside at 4 °C for 10 min, and then centrifuged at 13,000 rpm for 10 min at 4 °C. The supernatants from the two separation modes were filtered through 0.22 μm regenerated cellulose filters before HPLC–MS analysis. An in-house quality control (QC) was prepared by pooling and mixing the same volume of each sample. When devising the LC–MS analytical sequence, we randomly analyzed the samples to avoid any unrelated effects coming from systematic instrument changes.

2.4 Chromatographic

All chromatographic separations were performed using an Ultimate-3000 HPLC system (Dionex, USA). A 2.1 × 100 mm Kinetex C18 2.6 μm analysis column (Phenomenex, USA) along with a guard column (AQ0-8503, phenomenex, USA) was used for RP separation. The columns were maintained at 35 °C. The mobile phase was a mixture of (A) H_2O with 0.1 % FA and (B) ACN with 0.1 % FA, with a programmed gradient as follows: initial 15 % B maintained for 3 min, then increased to 50 % in 11 min, increased to 95 % in 4 min, held at 95 % for 10 min, decreased to 15 % in 0.1 min, and finally maintained at 15 % for 5 min. The injection volume was 15 μL, and the analysis column was eluted at a flow rate of 300 μL min^{-1}.

For HILIC analysis, a 2.1 × 150 mm XBridge™ HILIC 3.5 μm column (Waters, USA) was used on the same LC system. Mobile phase A consisted of 10 mM ammonium acetate, while mobile phase B was ACN modified with 0.1 % FA. The gradient started with an isocratic run at 95 % B for 3 min, decreased to 40 % in 15 min, maintained with 40 % in 10 min, and returned to 95 % for column equilibration for 8 min. The injection volume was 15 μL, and the analysis column was eluted at a flow rate of 200 μL min^{-1}.

2.5 Electrospray ionization mass spectrometry

A high resolution electrospray ionization (ESI) mass spectrometer (MicrOTOF-QII, Bruker Daltonics, USA) was operated in positive ion mode over the mass scan range of 50 to 1,000 to collect metabonomic profiles data. The acquisition rate was set at 1 spectrum per second. In the source, for RPLC mode the capillary voltage was set at −4,500 V, the end plate offset potential was set at −500 V, the nebulizer gas pressure was set at 0.8 bar, and the dry

gas flow rate was set at 7 L min^{-1} at a temperature of 200 °C. For HILIC mode the capillary voltage was set at −4,500 V, the end plate offset potential was set at −500 V, the nebulizer gas pressure was set at 0.7 bar, and the dry gas flow rate was set at 6 L min^{-1} at a temperature of 200 °C. A blank and a QC sample were injected every 7 samples to monitor the stability of the system.

2.6 Data extraction

All chromatograms were evaluated by Profile Analysis 1.1 software (Bruker Daltonics, Billerica, MA), which performed peak alignment, background noise subtraction, and data reduction in an automated and unbiased way. Only peaks with signal-to-noise ratio (S/N) greater than 5 were utilized in further analysis. All LC–MS detected peaks were identified by comparing both the MS spectra and the retention time. The main parameters were set as follows: retention time range 0.8–20 min for RPLC and 2.5–22 min for HILIC, mass range 100–1,000, mass window 0.5 Da, and retention window of 1 min.

2.7 Data normalization and statistical analysis

As is known to all, urine, unlike plasma, can have dramatic fluctuation in dilution concentration as a function of an individual's water intake and subsequent excretion. Normalization to total area of each sample is a more recently adopted technique and is an acceptable alternative to the method normalizing to urine creatinine and other normalization techniques (Cubbon et al. 2007; Clayton et al. 2006). One advantage of this normalization technique is that it can correct enrichment factors for different samples. Another advantage is that it can compensate inaccuracies caused by injection. This normalization technique was performed using ProfileAnalysis 1.1 software and the normalized data were exported into Excel (Microsoft) for further data manipulation.

Our previous study and other literatures have shown that reversed phase chromatography (RPLC) and hydrophilic interaction chromatography (HILIC) are complementary, and the combination of these two chromatographic techniques could obviously add information about metabolites to the study (Yin et al. 2009; Kind et al. 2007; Cubbon et al. 2010; Qin et al. 2008). In this study, both RPLC and HILIC have been coupled with a high resolution ESI mass spectrometer for a comprehensive urinary metabonomic profiling for each subject. The data sets from two columns were combined and analyzed by multivariate statistical analysis using SIMCA-P+ v12.0 software (Umetrics AB, Sweden). Principal component analysis (PCA) was employed to test the reproducibility of two separation modes separately. The combined dataset was subsequently analyzed by orthogonal partial least-squares-discriminant analysis (OPLS-DA), which can be applied to these complex spectral data to aid in the characterization of profile changes related to a biological perturbation (Bruce et al. 2008; Bylesjö et al. 2006). Each OPLS-DA model was evaluated by both internal permutation test and external validation test. Prior to PCA and OPLS-DA analysis, the profile data was mean-centered and unit variance scaled. Univariate ROC (receiver operator characteristic) analysis was carried out using software SPSS 17.0, to test the accuracy of each potential biomarker.

3 Results

3.1 Validation of analytical performance

Representative RPLC–MS and HILIC–MS total ion chromatograms (TICs) are shown in supplementary Fig. S1. The similar profiles indicated the difficulties of differencing BC and KC TICs by the naked eye. However, these differences are detectable by multivariate statistical analysis techniques. Research in metabonomics is critically dependent on the generation of sound analytical data (Gika et al. 2008). Thus, the concept of quality control (QC) samples reported by Gika et al. (2007) was adopted here to monitor and evaluate the stability of the analyses. The normalized data sets from RPLC and HILIC columns contained 6,786 and 2,716 variables, respectively. As shown in Fig. 1, 12 QC samples are tightly clustered in both separation modes, indicating the excellent stability and reproducibility of our chromatographic separation and mass measurement during the whole sequence. There are slight clustering trend between cancer patients and healthy controls. However, BC and KC patients cannot be distinguished at all. This is likely because unsupervised PCA technique separates samples without consideration of class information, and metabolite variations between BC and KC patients is smaller than the variations between cancer group and healthy controls. We also selected 9 representative peaks, which followed fair principle of covering a range of intensities and retention times, to verify the stability of the whole analyses (shown in supplementary Table S1).

3.2 Discriminate BC and KC patients from healthy controls

Cancer cells are known to possess a unique metabonomic phenotype, suggesting the possibility that OPLS-DA may be able to reveal cancer-specific patterns distinguishable from healthy ones, and be used to detect the presence of cancer (Issaq et al. 2008; Spratlin et al. 2009). The combined dataset form RP and HILIC modes contains 9561

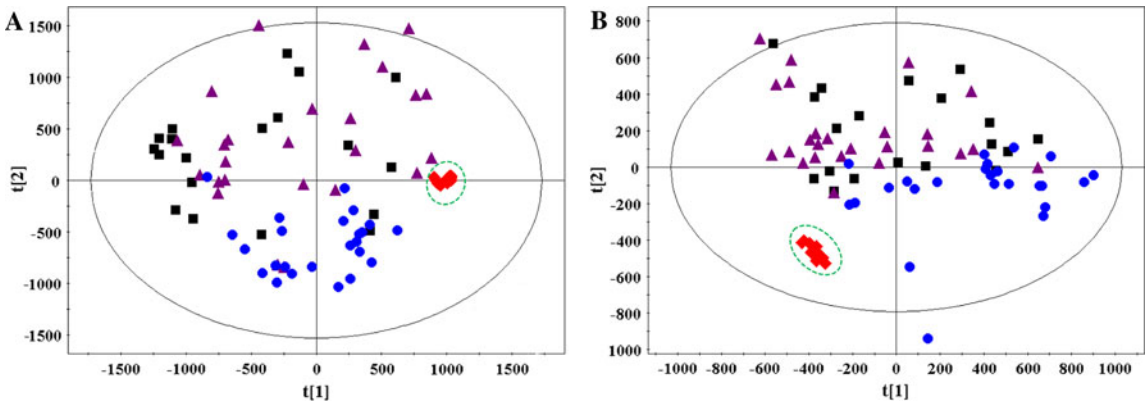


Fig. 1 PCA score plots based on the data from **a** RPLC and **b** HILIC separation. QC clusters (*filled diamond*) are highlighted within the *green ellipses*. BC (*filled square*), KC (*filled triangle*) patients and healthy controls (*filled circle*) are also colored (Color figure online)

variables. As expected, holistic OPLS-DA model based on the combined dataset is able to separate urine samples into two groups corresponding to BC patients and healthy controls (Fig. 2a). R^2X (cum) is 0.236, which is an important parameter to show how much data contributed to create the model. Two orthogonal components were calculated. Model parameters for the explained variation [R^2Y (cum)] and the predictive capability [Q^2 (cum)] are significantly high [R^2Y (cum) = 0.979; Q^2 (cum) = 0.681]. To guard against model overfitting, permutation tests with 100 iterations were carried out. The criteria for validity includes the following: all the permuted R^2 and Q^2 values to the left are lower than the original points to the right and the regression line of the Q^2 points intersects the vertical axis (on the left) at or below zero (Umetrics 2005). As shown in Fig. 2b, the validation plot strongly indicates that the model is valid. To evaluate the predictive ability of the OPLS-DA model, an external test using urine samples from 4 BC patients and 5 healthy controls was performed. None of the samples had been used in the generation of the model, which therefore allows for the estimation of true predictive accuracy. As shown in Fig. 2c, the OPLS-DA model correctly predicts all BC patients and all healthy controls. One attraction is that, 2 ES BC patients were also precisely predicted.

Application of holistic OPLS-DA resulted in clearly separation between KC patients and healthy controls, as shown in Fig. 2d. Two orthogonal components were calculated. The model parameters and validation plot suggest a good model (R^2X (cum) = 0.223; R^2Y (cum) = 0.957; Q^2 (cum) = 0.598; Fig. 2e). OPLS-DA class prediction was subsequently performed as for BC patients. As observed in Fig. 2f, the model correctly predicts 5 KC patients, in ES or advanced stages, and healthy controls, showing 100 % sensitivity and specificity. The loading plots for these two models are shown in supplementary Fig. S2.

3.3 Discriminate BC and KC groups

As described in the introduction section, it is necessary to investigate the clinical possibility of holistic metabonomic profiling on differencing BC and KC patients. Thus, we carried out the OPLS-DA based on 16 BC and 21 KC patients, while 3 BC and 4 KC patients were randomly selected as a test set. As depicted in Fig. 3a, BC patients are clearly separated from KC patients [R^2X (cum) = 0.201; R^2Y (cum) = 0.890; Q^2 (cum) = 0.467]. One orthogonal component was calculated. Validation plot obtained from 100 permutation tests confirmed the validity of this OPLS-DA model (data not shown). All test subjects, including 4 ES cancer patients (2 each of BC and KC), are correctly predicted without crossover (Fig. 3b). This result indicates the feasibility of holistic metabonomic profiling of urine in discriminating BC and KC in both ES and advanced stages.

S-plot derived from OPLS-DA model comparing BC to KC is shown in Fig. 3c. The S-plot is a visualized method to interpret metabonomic data. By the combination of covariance (*X*-axis) and correlation loading profiles (*Y*-axis), the S-plot could thus be helpful for filtering interesting variables (metabolites) and lowering false positives. The variables with higher *P* and *P*(corr) values were the most relevant metabolites for the separation. Thus, those beyond the threshold (in two shaded areas) indicate the most reliable variables that greatly influence the separation in Fig. 3a (Wiklund et al. 2008). Variables without the support of necessary confidence intervals were rejected (Yin et al. 2009). As shown in Fig. 3d, two variables highlighted in red could not be considered of significance because of their negative confidence intervals (Yin et al. 2009). The qualified variables that meet above criteria are the best possible biomarkers for BC or KC, as shown in Table 2. Unpaired Student's *t* tests were performed in succession. Eleven variables are significantly different in concentration

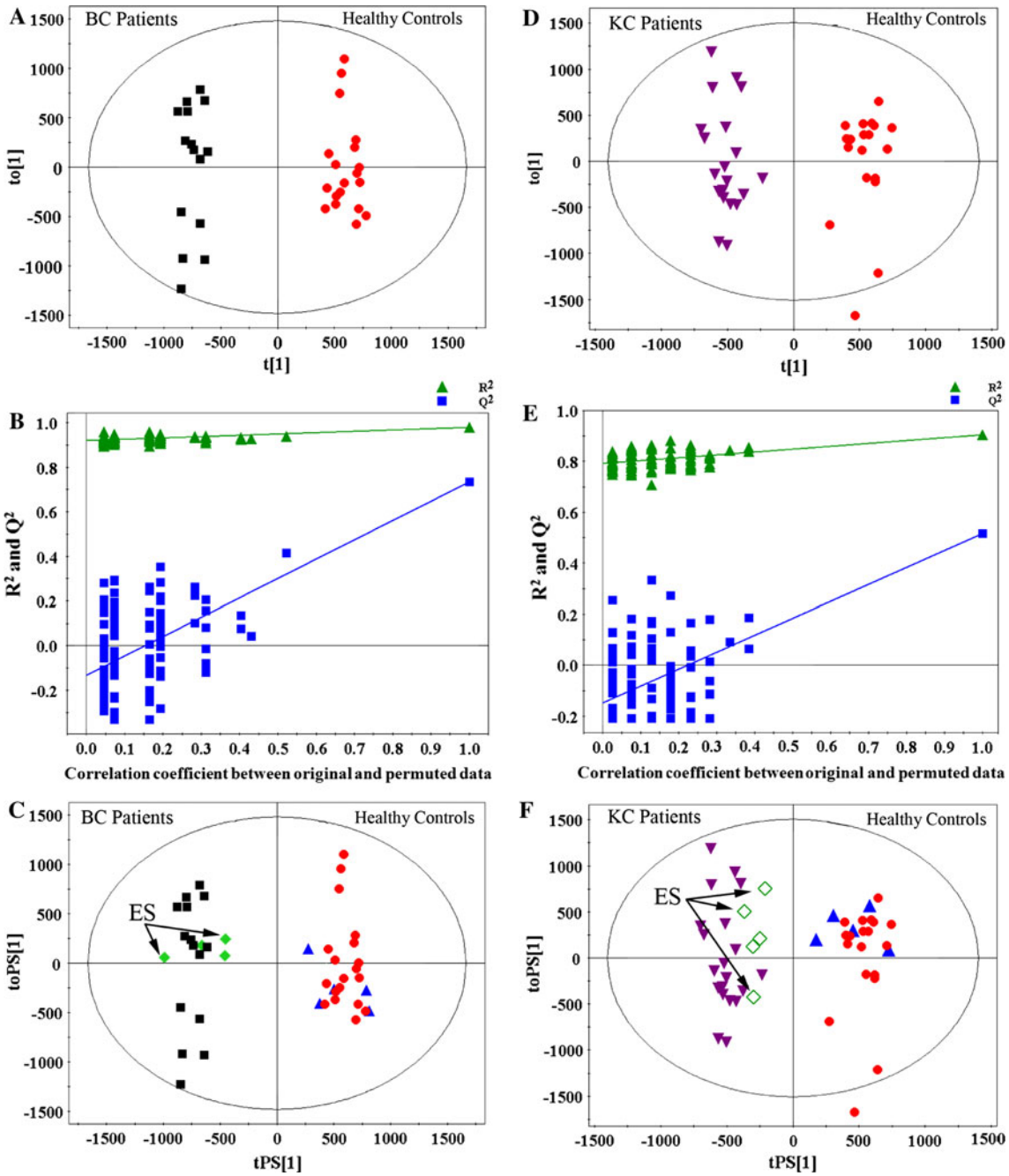


Fig. 2 Urinary metabonomic profiles derived from BC and KC patients are different from healthy controls. **a** OPLS-DA model with 15 BC patients (*filled square*) and 19 healthy controls (*filled circle*). **b** Permutation validation of the OPLS-DA model in Fig. 2a. R^2 is the explained variance, and Q^2 is the predictive ability of the model. **c** OPLS-DA prediction of 4 additional BC patients (*filled diamond*) and 5 additional healthy controls (*filled triangle*). **d** OPLS-DA model with 20 KC patients (*filled inverted triangle*) and 19 healthy controls (*filled circle*). **e** Permutation validation of the OPLS-DA model in Fig. 2d. **f** OPLS-DA prediction of 5 additional KC patients (*open diamond*) and 5 additional healthy controls (*filled triangle*). ES patients were specifically *arrowed*

between KC and BC, with P value <0.05. The detailed method for metabolite identification has been described previously (Lin et al. 2011; Kind and Fiehn 2010), and it is further illustrated in supplementary Fig. S3 for this study.

Many metabolites in Table 2 demonstrate relatively decreased concentration levels with a cancer phenotype (KC and BC) when compared with the healthy controls. More importantly, the extent of the change is different between BC and KC group. For example, based on VIP

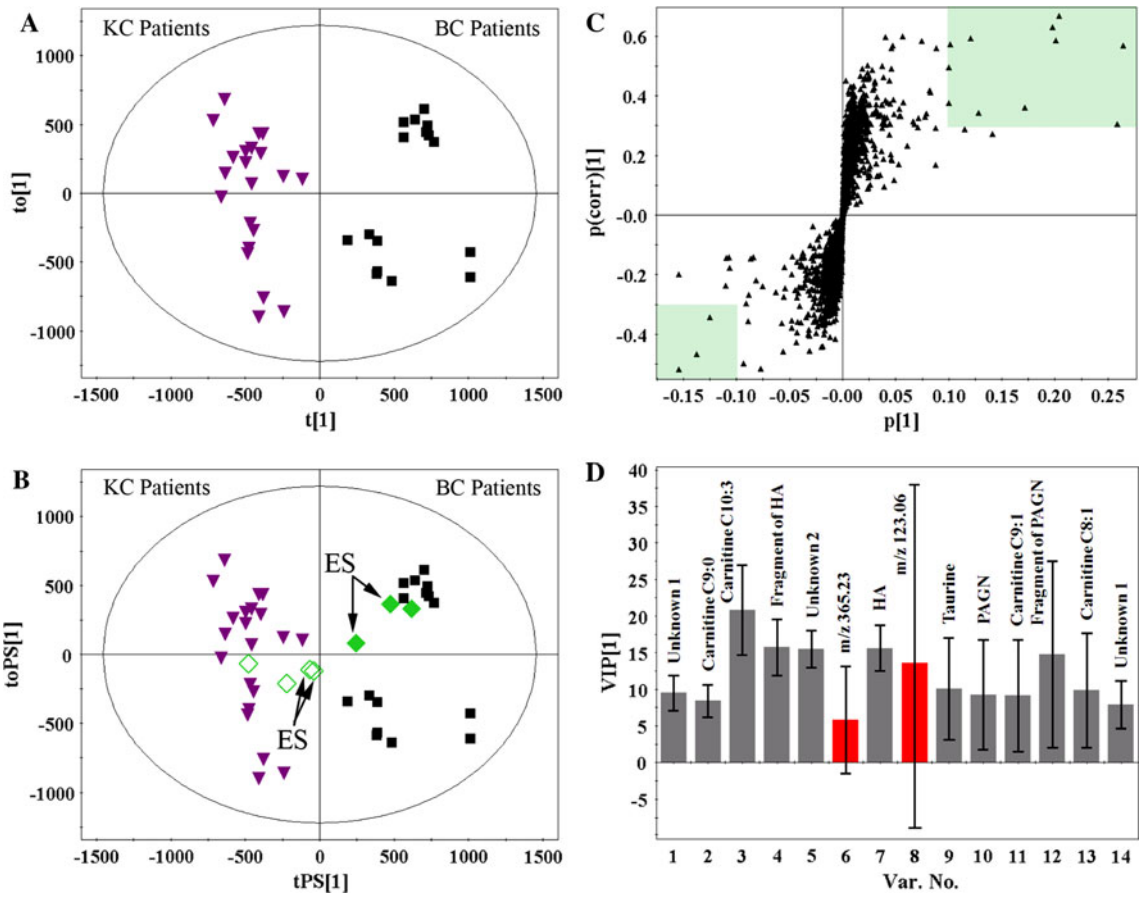


Fig. 3 Urinary metabonomic profiles derived from patients with BC and KC are different. **a** OPLS-DA model comparing 16 BC (*filled square*) with 21 KC patients (*filled inverted triangle*). **b** OPLS-DA prediction of 3 additional BC patients (*filled diamond*) and 4 additional KC patients (*open diamond*). **c** S-plot at the first component used in potential biomarkers selection. **d** VIP column plot with jack-knifed confidence intervals. Variables 6 and 8 were eliminated because of their negative confidence intervals. ES patients were specifically *arrowed*

(Variable Importance Project) value, HA is ranked as the most important variable responsible for separating BC patients, with a 90 % decrease in concentration relative to healthy controls. However, this variable's importance drops to 5th for KC with a 40 % decrease in concentration. Carnitine C10:3 is as the most important metabolite for discriminating KC patients, with a 30 % decrease in concentration relative to healthy controls. Nevertheless, its importance falls to 10th with a 20 % increase in concentration for BC. The altering degrees of metabolite concentration reflect the intricate nature of each cancer. Through metabonomic classification, we are able to attribute the different extents of concentration change to BC and KC. The low variation of biomarker concentrations in QC samples confirmed that the alterations of those metabolites in our study were due to biological changes rather than analytical errors (Table 2).

Diagnostic accuracy is usually evaluated using the area under the ROC curve (AUC), which combines the sensitivity and specificity of a given biomarker for the diagnosis of a specific disease (Poynard et al. 2007). Here, univariate ROC analysis was carried out for each variable with their results AUC displayed in Table 2. All subjects in the training set and predictive set were used in this test. For those identified potential biomarkers, HA and taurine are the most likely biomarkers for BC, because significantly difference ($P < 0.05$) was observed when comparing these two metabolites in BC group to both healthy controls and KC groups. HA performs the best because its AUC reaches 0.855 (AUC < 0.5 is considered no discrimination power). Ideal disease biomarkers should exhibit high sensitivity and high specificity (Poynard et al. 2007). For BC diagnosis, HA achieves 78.9 and 86.5 % for sensitivity and specificity, respectively (Fig. 4a). Likewise, carnitine C10:3 is of significance in separating KC from both BC patients and healthy controls, and is considered the most possible biomarker for KC diagnosis. Its AUC is 0.713, as shown in Table 2. Carnitine C10:3 reaches 84.0 % for sensitivity and 60.5 % for specificity (Fig. 4b).

Finally, holistic OPLS-DA model was built based on the combined data from 15 BC patients, 21 KC patients and 20 healthy controls. Comparison of the three groups revealed

Table 2 Metabolites contributed the most in BC and KC classification

m/z	Metabolite	BC versus KC			BC versus healthy controls			KC versus healthy controls			AUC[c]		% RSD[d]
		Ratio[a]	VIP	*P*[b]	Ratio[a]	VIP	*P*[b]	Ratio[a]	VIP	*P*[b]	BC	KC	
310.20	Carnitine C10:3[e]	1.7	21.2	6.7E−03	1.2	3.8	<0.05	0.7	10.5	4.5E−02	<0.5	0.713	4.4
105.03	Fragment of HA[f]	0.2	16.8	2.8E−03	0.2	13.8	2.8E−07	0.7	5.1	<0.05	0.854	<0.5	5.6
180.07	HA[f]	0.1	16.7	6.2E−03	0.1	13.3	1.9E−06	0.6	6.2	<0.05	0.855	<0.5	2.9
358.25	Unknown 2	2.9	15.1	4.5E−02	0.9	1.9	<0.05	0.3	10.1	1.2E−04	<0.5	0.743	9.4
130.05	Fragment of PAGN[f]	0.6	14.6	3.2E−02	0.6	8.8	2.9E−03	0.9	3.2	<0.05	0.728	<0.5	3.5
286.19	Carnitine C8:1[e]	1.6	9.8	2.4E−02	1.5	7.5	<0.05	0.9	6.3	<0.05	<0.5	<0.5	4.1
126.01	Taurine[e]	2.7	9.8	1.1E−04	2.5	6.3	4.1E−04	0.9	<1	<0.05	0.773	<0.5	10.3
348.10	Unknown 1	14.8	9.2	1.7E−05	20.7	9.6	2.4E−05	1.4	<1	<0.05	0.853	<0.5	6.9
265.11	PAGN[f]	0.6	8.5	<0.05	0.5	11.9	5.0E−03	0.8	4.3	<0.05	0.734	<0.5	2.1
300.20	Carnitine C9:1[e]	0.7	8.3	<0.05	0.3	8.5	5.2E−04	0.5	5.8	1.2E−02	0.675	0.684	6.1
302.23	Carnitine C9:0[e]	0.8	7.9	<0.05	0.2	9.7	1.0E−04	0.3	9.3	3.0E−04	0.673	0.687	14.8

HA hippuric acid, *PAGN* phenylacetylglutamine

[a] Ratio was defined as peak area ratio, and calculated from the arithmetic mean values of each group

[b] *P* value calculated using unpaired Student's *t* test

[c] AUC was used to evaluate the clinical importance of each metabolite

[d] Variation of the metabolite concentrations in QC samples expressed as relative standard deviation (% RSD)

[e] Metabolites formally validated by standard samples

[f] Metabolite identified based on database information or from the literature

clear class distinction with corresponding R^2X (cum) = 0.187, R^2Y (cum) = 0.894 and Q^2 (cum) = 0.491 (Fig. 4c). Two orthogonal components were calculated. Permutation tests with 100 iterations confirmed that the model was valid and the separation was not due to chance correlation (the plot is not shown). OPLS-DA class prediction was performed on 12 subjects, which had not been used in the generation of the model. As observed in Fig. 4d, all subjects are correctly predicted. More importantly, ES cancer patients are precisely predicted as well. The metabolites most significant for this three-class separation are HA, PAGN, carnitine C9:1, carnitine C8:1, carnitine C10:3, proline betain, taurine and carnitine. Comparing to metabolites causing separation in the BC versus KC model, proline betain and carnitine were newly observed while carnitine C9:0 was not found. AUC of proline betain for discriminating BC was calculated 0.785, and for KC was smaller than 0.5. AUC of carnitine for discriminating KC was calculated 0.586, and for BC was smaller than 0.5. From the results of AUC, HA and Carnitine C10:3 are still the most possible biomarkers for BC and KC, respectively. These results indicate that holistic metabonomic profiling in conjunction with OPLS-DA based on multiple-metabolites are more powerful than current single metabolite methods. We thus advocate the holistic metabonomic profiling of urine as a noninvasive tool to diagnose BC and KC.

4 Discussion

Among the metabonomic surveys of human cancer, BC and KC are topics of great concern (Issaq et al. 2008; Kim et al. 2009; Pasikanti et al. 2010; Lin et al. 2011). Pasikanti et al. (2010) used GC–MS for urinary metabonomic profiling for BC patients and non-BC controls, and demonstrated that it was amenable to the noninvasive diagnosis of human BC based on all qualified profile data. They also identified marker metabolites including senecioic acid, 2-butenedioic acid, ribonic acid, 2,5-furandicarboxylic acid, melibiose, sumiki's acid, uridine, 2-propenoic acid, glycerol, gluconic acid, valerate, fructose, L-valine, citric acid, ribitol, that accounted for the differentiation of BC and non-BC cohorts. Unfortunately, they did not further confirm the diagnosis accuracy of these biomarkers. Issaq et al. (2008) reported the feasibility of holistic metabonomic profiling of urine by RPLC–MS to discriminate between controls and BC patients. But, potential marker metabolites responsible for the separation were not further identified (Issaq et al. 2008). In general, for BC, the accuracy of previously reported urinary marker metabolites have not been discussed yet. For KC, Kim et al. (2009) have shown that examining the entire metabolome in the urine (including known as well as unknown) can result in the separation from non-malignancy controls. Regrettably, they did not enroll BC in their control set, and the ability of

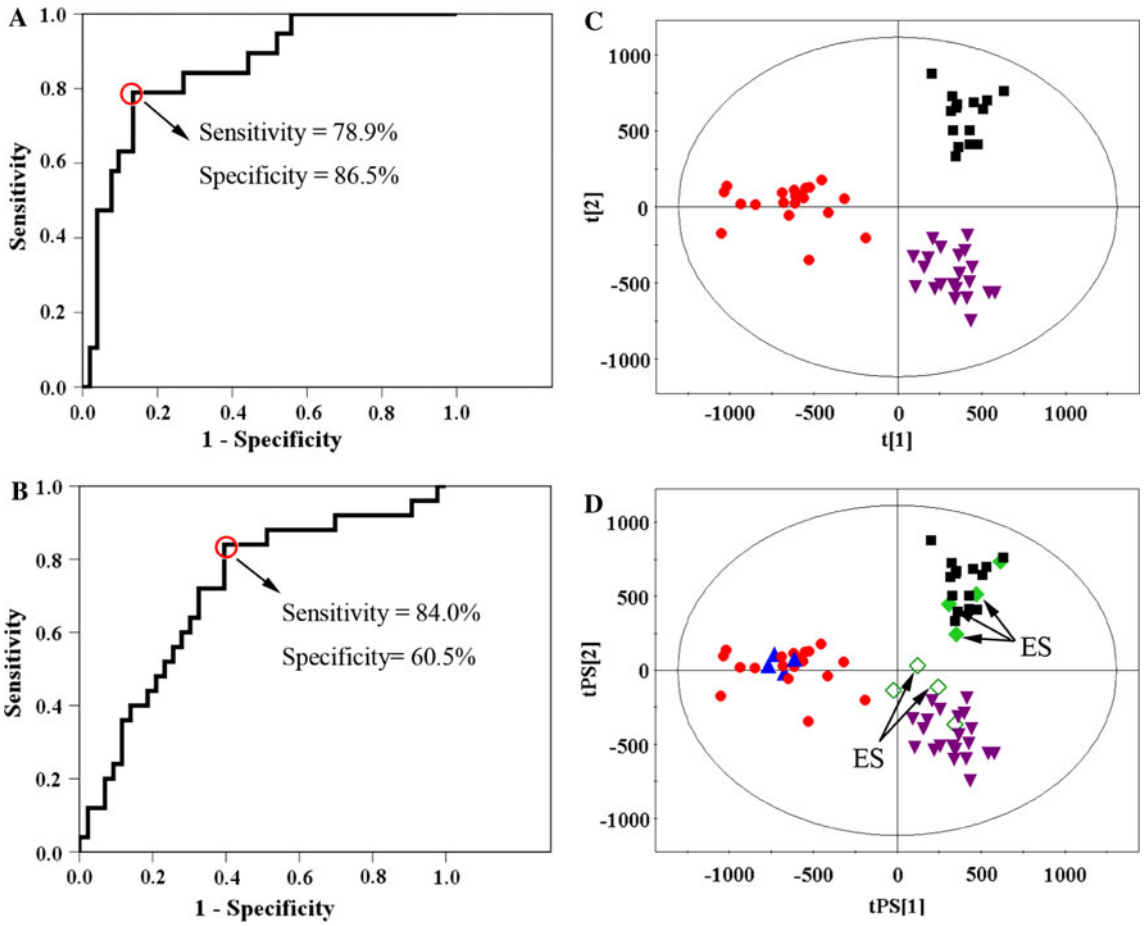


Fig. 4 **a** Diagnostic efficacy evaluation using ROC curve for HA. **b** Diagnostic efficacy evaluation using ROC curve for carnitine C10:3. The *open circles* denote best cut-off points. **c** OPLS-DA model built based on data from BC (*filled square*), KC (*filled inverted triangle*) patients and healthy controls (*filled circle*). The model clearly separated these three groups without any overlaps. **d** OPLS-DA prediction of 12 additional subjects including 4 BC patients (*filled diamond*), 4 KC patients (*open diamond*) and 4 healthy controls (*filled triangle*). ES patients were specifically *arrowed*

metabonomic profiling on BC and KC discrimination has not been demonstrated.

HA is a carboxylic acid found in urine, and its excretion is associated with environmental-toxic exposures of aromatic compounds, or dietary protein degradation and re-synthesis (Gamagedara et al. 2012). According to our study, urine HA concentration shows pronounced decrease in BC group, which seems unique to BC. Yang et al. (2010) showed that urine HA were 1.9-fold higher in lung cancer than that in healthy controls. Catchpole et al. (2011) found HA concentration level were greatly decreased in KC tissues, and the decrease contributed most in the KC and healthy discrimination. Unfortunately, they did not demonstrate whether urine HA was expressed abnormally.

As shown in Table 2, carnitine C9:0 and carnitine C9:1 are lower expressed in BC and KC groups than those of healthy controls, but neither unique for BC nor KC. It is the possible result of any malignancy. Reduced levels of those two metabolites in urine were also found in liver cancer patients (Chen et al. 2009). Both carnitine C9:0 and carnitine C9:1 function as important intermediates in fatty acid transport across mitochondrial membranes. A small amount of them escapes from mitochondria and then be secreted into urine. McClinton et al. (1991) raised one possibility, that is, development of tumor might result in dysregulation of lipid metabolism. This assumption, in some extent, could explain why urine carnitine C9:0 and carnitine C9:1 are indistinguishably expressed in BC and KC groups, but significantly different from healthy controls. Interestingly, carnitine C10:3 are specific down-regulated in KC patients. The similar down-regulation was also observed in liver cancer patients (Chen et al. 2009). Taurine is 2.5-fold higher in BC group than that in healthy controls in our study, while shows little difference when compared KC to healthy controls. Srivastavaa et al. (2010) also evidenced it as a possible fingerprint biomarker for non-muscle invasive BC when take healthy subjects as controls. However, elevated taurine levels were detected in malignant breast and brain tissues,

colon mucosa, and lung cancer urine (Beckonert et al. 2003; Moreno et al. 1993). Taurine has long been discussed as an osmoregulator and a biomarker of increasing cell proliferation or a potential indicator of tumor aggressiveness. Strangely, in our study, there are no significant concentration changes of urine taurine in KC group, though 64 % of KC patients were in advanced-stage.

It is appropriate to be curious why massive efforts to seek specific metabolic biomarkers for BC and KC have made no satisfactory discoveries until now, and why holistic urinary metabonomic profiling can discriminate them precisely. Potential metabolic biomarkers reported previously have a common characteristic of high abundance in biofluids (Yin et al. 2009; An et al. 2010; Lin et al. 2011). As a result, those biomarker-screening methods may easily be swamped by these abundant metabolites and neglect detectable amounts of the low-abundance ones. In contrast, holistic metabonomic profiling with OPLS-DA does not write off the contribution of low-abundance metabolites in classification. As for BC and KC, detectable single biomarker usually cannot satisfy sensitivity and specificity requirements at the same time (as shown in Table 2; Fig. 4). Holistic profiling methods as OPLS-DA modeling utilizes all information, therefore, can detect more specific BC and KC biomarker patterns, and offers greater potential in disease diagnosis. More significantly, BC and KC are extremely complex tumorigenesis processes, which involve large amounts of metabolites, not limited to just one or two metabolites (Evan and Vousden 2001). Diagnostic approaches based on limited metabonomic biomarkers, to some extent, are less efficient than multiple-biomarkers methods. Thus, the development of holistic approaches could be advocated (Yin et al. 2009).

In addition to the accurate diagnosis, staging of cancer is significant for cancer management and critical in predicting prognosis. One factor to note is that a large number of cancer patients in our study (73.7 % of BC and 36 % of KC) are in ES, and that they were impartially classified by holistic metabonomic profiling of urine. This adds to the evidence supporting of using the holistic profiling as an effective method to detect BC and KC in ES.

5 Conclusions

In summary, we have addressed issues of urine component change due to bladder and kidney tumorigenesis. The results suggest that urinary holistic metabonomic profiling could be more accurate for BC and KC diagnosis than just one or two metabonomic biomarkers, has the diagnosis ability for BC and KC in ES, and can discriminate these top-two-incidence genitourinary cancers from each other. For its noninvasive, efficient, and low-cost features, we thus recommend the holistic urinary metabonomic profiling with OPLS-DA be an alternative prescreen technique to other forms of screening methods for BC and KC.

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